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3.3.4 Identity-4: $(a + b)(a - b) = a^2 - b^2$

In the given figure, AB = AD = a.

So, area of square ABCD = a^2.

Also, SB = DP = b.

Area of the rectangle SBCT = ab.

Similarly, area of the rectangle DPRC = ab.

Also, area of the square TQRC = b^2.

Area of the rectangle DPQT = $ab - b^2$.

Now, AS = PQ = $(a - b)$ and AP = SQ = $(a + b)$.

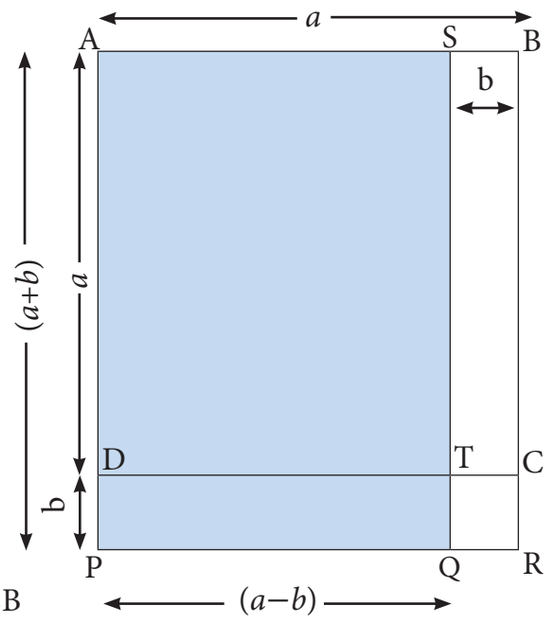


Fig. 3.17

Hence, area of the rectangle APQS (the shaded rectangle) = area of square ABCD − area of rectangle STCB + area of rectangle DPQT

$= a^2 - ab + (ab - b^2)$

$= a^2 - ab + ab - b^2$

$= a^2 - b^2$

Hence, $(a + b)(a - b) = a^2 - b^2$

Q2Y3L8

Example 3.4 Simplify by using the identity $(a + b)(a - b) = a^2 - b^2$.

(i) $(3x + 4)(3x - 4)$ (ii) 53×47.

Solution

(i) $(3x + 4)(3x - 4)$

Substitute, $a = 3x$ and $b = 4$ in the identity $(a+b)\times(a-b) = a^2 - b^2$, we get,

$(3x + 4)(3x - 4) = (3x)^2 - 4^2$

$(3^2 \times x^2) - 16 = 9x^2 - 16$.

(ii) $53 \times 47 = (50 + 3) \times (50 - 3)$.

Take, $a = 50$ and $b = 3$,

Substituting the values of 'a' and 'b' in the identity $(a + b)(a - b) = a^2 - b^2$), we get,

$= 50^2 - 3^2$

$= 2500 - 9$

$= 2491$.

Try this

Consider a square shaped paddy field with side of 48 *m*. A pathway with uniform breadth is surrounded the square field and the length of the outer side is 52 *m*. Can you find the area of the pathway by using identities?

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